



# Mingdong Li

Last updated: November 17, 2025

🎓 Google Scholar 🌐 Homepage: [mingdong-li.github.io](https://mingdong-li.github.io)  
✉️ [mlidr@connect.ust.hk](mailto:mlidr@connect.ust.hk)  
☎️ +86 15221361908 📞 +852 62059030

## About Me

My research interest lies in Brain-machine Interface, NeuroAI, Cognitive Analysis and Robotics. During my PhD study, I focused on point processes modeling (spiking activity) for understanding neural populations coding in Brain-machine Interfaces. Now I am designing algorithms for tactile sensors and robotics manipulation.

## Education

|                   |                                                                                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021.09 – 2025.08 | <b>Hong Kong University of Science and Technology</b><br>Ph.D in Robotics and Autonomous Systems (IIP Program)<br>Thesis: Modeling Functional Neural Connectivity in Point Process Filtering for Brain-machine Interface<br>Supervisor: Prof. Yiwen Wang and Prof. Qifeng Chen (co-supervisor) |
| 2018.09 – 2021.04 | <b>Zhejiang University</b> , MPhil. in Mechanical Engineering                                                                                                                                                                                                                                  |
| 2013.09 – 2018.07 | <b>Tongji University</b> , B.Sc. in Mechanical Engineering                                                                                                                                                                                                                                     |

## Experience

|                                                                                                                                                                                                                                            |                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Anker Innovation Inc.</b> <i>Dexterous Hand Lab, Algorithm Researcher</i>                                                                                                                                                               | 2025.04 – Present |
| <ul style="list-style-type: none"><li>• Tactile sensor survey and planning</li><li>• Tactile sensor representation and robotics manipulation algorithm</li><li>• LangGraph-based AI tool development for robotics talent seeking</li></ul> |                   |

## Publications

[Working Papers]:

1. **Mingdong Li**, Zhiwei Song, Shuhang Chen, Xiang Zhang, Yiwen Wang\*. Dynamic Functional Neural Connectivity Inference for Multi-Task Neuroprosthetic Control in a Point Process Filter, ***IEEE Transactions on Biomedical Engineering (TBME)***. (under revision)

[Journals]:

1. Shenghui Wu, Zhiwei Song, Xiang Zhang, Yifan Huang, Shuhang Chen, Xiang Shen, Jieyuan Tan, **Mingdong Li**, Ziyi Wang, Yujun Chen, Kai Liu, Dario Farina, Jose Principe, Yiwen Wang\*. Re-establishing neural functional connectivity with a generative spike prediction model using behavioral reinforcement, ***Nature Computational Science***, 2025.
2. Zhiwei Song, Xiang Zhang, **Mingdong Li**, Jieyuan Tan, Yiwen Wang\*. An Online Knowledge Transfer Framework for Task Learning in Brain-Machine Interfaces, ***IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)***, 2025.
3. **Mingdong Li**, Shuhang Chen†, Xiang Zhang, Yiwen Wang\*. Neural Correlation Integrated Adaptive Point Process Filtering on Population Spike Trains, ***IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)***, 2025. (†: co-first author)
4. **Mingdong Li**, Shanhe Lou\*, Hao Zheng, Yixiong Feng, Yicong Gao, Siyuan Zeng, Jianrong

Tan. A Cognitive Analysis-based Key Concepts Derivation Approach for Product Design, **Expert Systems With Applications (ESWA)**, 2024.

5. **Mingdong Li**, Shanhe Lou\*, Yicong Gao, Hao Zheng, Bingtao Hu, Jianrong Tan. A Cerebellar Operant Conditioning-inspired Constraint Satisfaction Approach for Product Design Concept Generation, **International Journal of Production Research (IJPR)**, 2023.
6. Xuanyu Wu, Zhaoxi Hong\*, Yixiong Feng, **Mingdong Li**, Shanhe Lou, Jianrong Tan. A Semantic Analysis-driven Customer Requirements Mining Method for Product Conceptual Design, **Scientific Reports (Sci. Rep.)**, 2022.
7. Yixiong Feng, **Mingdong Li**, Shanhe Lou\*, Yicong Gao, Jianrong Tan. A Digital Twin-Driven Method for Product Performance Evaluation Based on Intelligent Psycho-Physiological Analysis, **ASME Journal of Computing and Information Science in Engineering (JCISE)**, 2021.

[Conferences & Patents]:

1. **Mingdong Li**, Tianyi Hu, Zhiwei Song, Mingyi Wang, Shenghui Wu, Jieyuan Tan, Zixu Wang, Yiwen Wang\*. An Estimating Neural Representations Underlying Behaviors via Point Process Filtering in Brain-machine Interface Task Learning, **IEEE EMBS 12th Annual International Conference on Neural Engineering (NER)**
2. **Mingdong Li**, Mingyi Wang, Yiwen Wang\*. An Adaptive Superposition Point Process Model with Neuronal Encoding Engagement Identification, **2024 46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)**. (Oral)
3. Zixu Wang, Shuhang Chen†, **Mingdong Li**, Yiwen Wang\*. Tracking Dynamic Conditional Neural Correlation during Task Learning, **2024 46th EMBC**. (Oral, †: co-first author)
4. **Mingdong Li**, Shuhang Chen, Zhijia Zhao, Yiwen Wang\*. Tracking the Dynamic Functional Neural Connectivity via Conjugate Gradient Optimization, **2023 45th EMBC**. (Oral)
5. **Mingdong Li**, Shuhang Chen, Xi Liu, Zhiwei Song, Yiwen Wang\*. Modeling Neural Connectivity in a Point-Process Analogue of Kalman Filter, **2022 44th EMBC**.
6. Y. Feng, **Mingdong Li**, Y. Gao. CN110090818B, Chinese Patent Authorized, 2020
7. M. Li, Y. Zhang, N. Xu, **Mingdong Li**, S. Liu. CN106127958B, Chinese Patent Authorized, 2018

## Award

- |                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| 1. HKUST ECE Future Leader                                                         | 2025      |
| 2. German Academic Exchange Service (DAAD) Alnet fellowship (AI4Science)           | 2024      |
| 3. NextGen Scholar Award (IEEE Annual International Conference of EMBS)            | 2024      |
| 4. Zhejiang University Dissertation-Year Fellowship (Top 1%)                       | 2021      |
| 5. ZHEJIANG Lab AI Competition (Multiple Objects Tracking track), Excellence Prize | 2019      |
| 6. 1st prize in No.7 National College Mechanical Design Innovation Competition     | 2016      |
| 7. 1st prize in Shanghai College Mechanical Innovation Competition                 | 2016      |
| 8. Tongji Academic Scholarship, 2nd prize for 2 times and 3rd prize for 1 time     | 2013-2018 |

## Service

- Reviewer:
  - Journal of Neural Engineering (JNE), IEEE Transaction on Intelligent Vehicles (TIV)
  - Conference: EMBC, ISBI, NER
- Teaching Assistant
  - EMIA4110 Practical Machine Learning

2024 Spring

- ELEC4130 Machine Learning on Images
- Program Committee
  - 4th International Workshop on Neural Engineering & Rehabilitation
  - 3rd International Workshop on Neural Engineering & Rehabilitation

## Skills

---

- **Coding:** Python/Matlab/C++/Pytorch toolbox
- **Product Design:** AutoCAD/Inventor/Solidworks/Ansys
- **Language:** Chinese, English, German
- Surgery for implementing electrodes into M1 and mPFC of SD Rat Establish system for animal behavioral training, neural signal recording and analysis